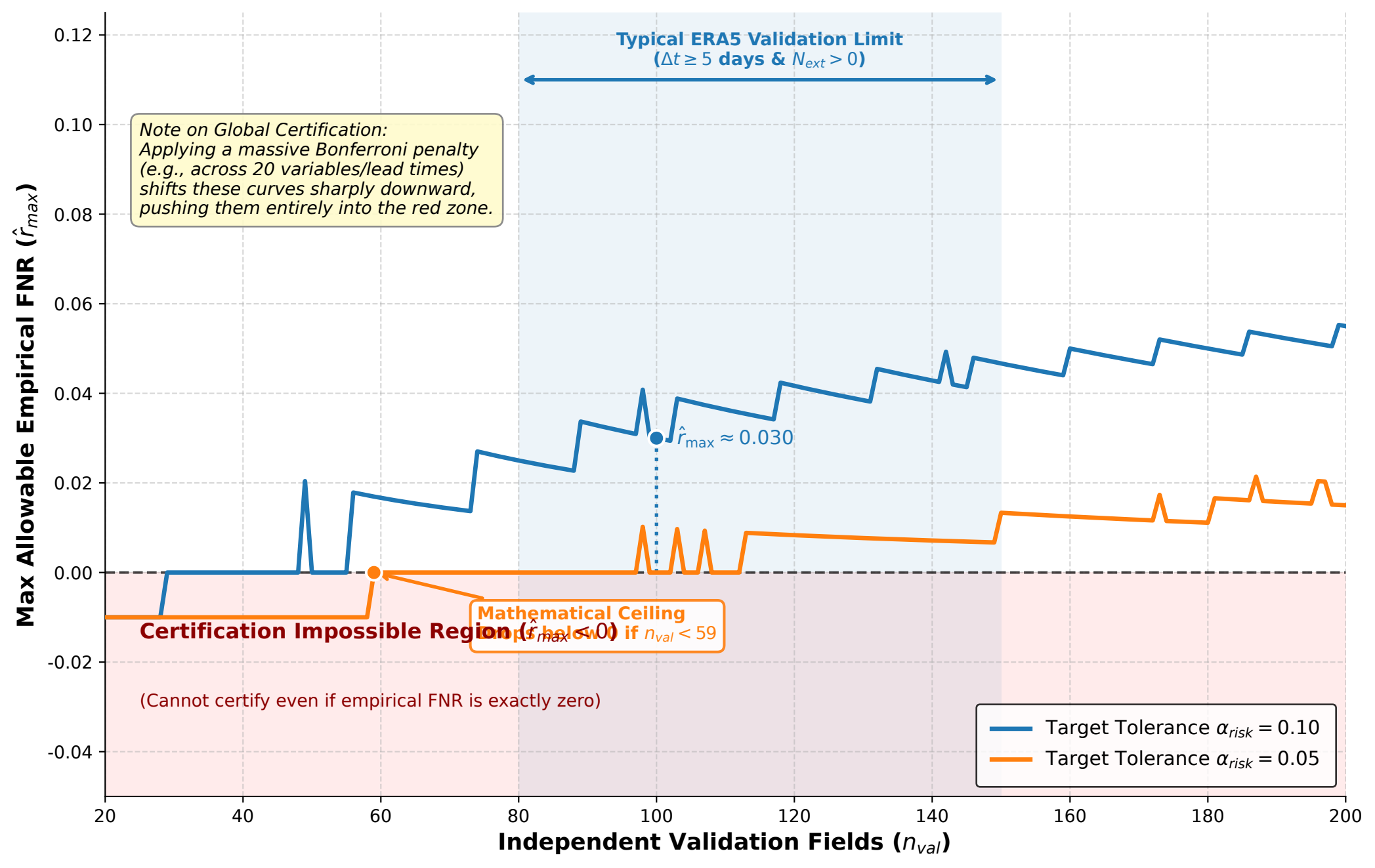
Typical ERA5 Validation Limit
($\Delta t \geq 5$ days & $N_{ext} > 0$)
Note on Global Certification:
Applying a massive Bonferroni penalty
(e.g., across 20 variables/lead times)
shifts these curves sharply downward,
pushing them entirely into the red zone.
$\hat{r}_{max} \approx 0.030$
Mathematical Ceiling
Impossible if $n_{val} < 59$
Certification Impossible Region ($\hat{r}_{max} < 0$)
(Cannot certify even if empirical FNR is exactly zero)
Target Tolerance $\alpha_{risk} = 0.10$
Target Tolerance $\alpha_{risk} = 0.05$
Max Allowable Empirical FNR ($\hat{r}_{max}$)
Independent Validation Fields (n_{val})
0.12
0.10
0.08
0.06
0.04
0.02
0.00
-0.02
-0.04
20
40
60
80
100
120
140
160
180
200